

SDC Assignment 5 - Application of EKF

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1 Results

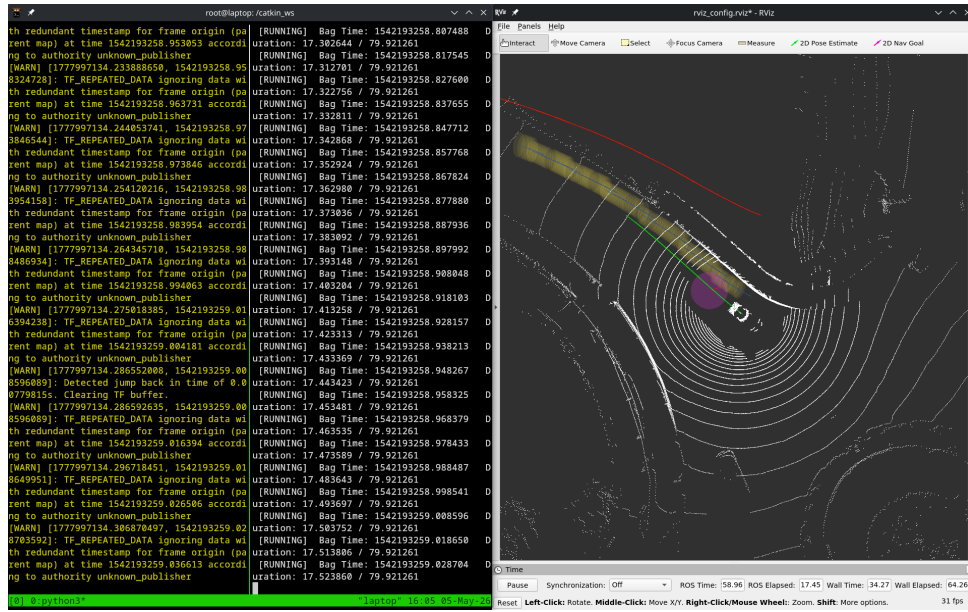


Figure 1: The screenshot of the running program.

YouTube Link: <https://youtu.be/oK6bKV5-J7w>

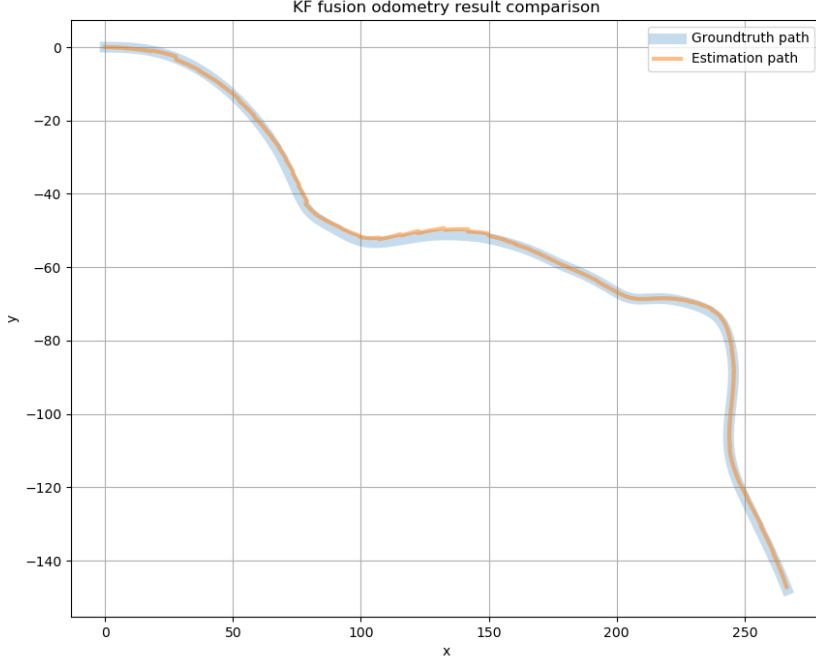


Figure 2: The result of the EKF.

2 Discussion

2.1 The design of the EKF

The Extended Kalman Filter (EKF) is designed to estimate a 3D state vector $[x, y, yaw]^T$. The design consists of two main steps:

- **Prediction Step (Radar Odometry):** Uses a non-linear motion model updating the pose based on local control inputs dx , dy , and $dyaw$. The state transition matrix A is the Jacobian of this model. The state covariance S is updated using A and the process noise covariance R .
- **Update Step (GPS):** Uses an observation matrix C mapping the 3D state to a 2D $[x, y]^T$ measurement. The Kalman Gain K is computed using C , S , and the measurement noise covariance Q . Finally, the state and covariance S are corrected based on the measurement residual.

2.2 The covariance matrix of GPS, radar odometry

Definition:

- Both sensor messages provide a 6×6 covariance matrix.
- **Radar Odometry:** A 3×3 sub-matrix for x , y , and yaw is extracted and assigned as the process noise covariance R .
- **GPS:** A 2×2 sub-matrix for x and y is extracted and assigned as the measurement noise covariance Q .

Meaning: The covariance matrix represents the expected noise or uncertainty of the sensor data. A higher variance indicates lower certainty. By comparing the radar odometry covariance (R) and GPS covariance (Q), the Kalman Gain determines whether the filter should rely more on the physical motion prediction or the absolute GPS measurement.